

I) M1: $I_{ref} = \frac{1}{2} k' \left(\frac{W}{L}\right)_1 (V_{G_S} - V_{th})^2 \left(1 + \lambda V_{G_S}\right)$ Aprox. = 0

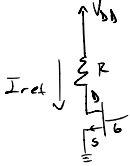
$$60 \mu A = \frac{1}{2} (120 \frac{\mu A}{V^2}) (5) (V_{G_S} - 0.6V)^2$$

$$60 [\mu A] \cdot \frac{2}{5} \cdot \frac{1}{120} \left[\frac{V^2}{\mu A}\right] = (V_{G_S} - 0.6V)^2$$

$$0.2 [V^2] = (V_{G_S} - 0.6V)^2 \quad \sqrt{\quad}$$

$$0.44 [V] = V_{G_S} - 0.6V$$

$$1.047 = V_{G_S}$$



$$I_{ref} = \frac{V_{DD} - V_{G_S}}{R}$$

$$R = \frac{3.3 - 1.047}{60 \mu A} = 37550$$

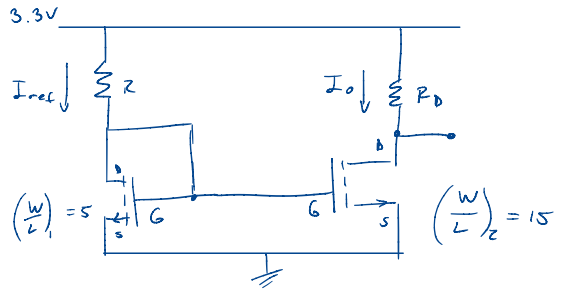
II) $I_{O \text{ ideal}} = I_{ref} \frac{(W/L)_2}{(W/L)_1} = 60 \mu A \cdot \frac{15}{5} = 180 \mu A$

$$I_{O \text{ real}} = I_{ref} \frac{(W/L)_2}{(W/L)_1} \cdot \frac{(1 + \lambda V_{DS2})}{(1 + \lambda V_{DS1})}$$

$$= 180 \mu A \cdot \frac{(1 + (0.025)(3.3 - 10 \mu A I_O))}{1.026}$$

$$I_O = \frac{180 \mu A}{1.026} (1.08 - 250 I_O)$$

$$I_O = 181.93 \mu A$$



$$k' = 120 \mu A/V^2, V_{th} = 0.6V, \lambda = 0.025 V^{-1}$$

$\hookrightarrow$ Transconductancia $\approx$ Umbral para $\nearrow$ Variaciones en V_{DS}
 activar $\nearrow$ afectan corriente (efecto early)

$k' = \mu_n \cdot C_{ox}$
 $\nearrow$ Capacitancia por unidad de área
 movilidad de e^-

1) $I_{ref} = 60 \mu A, R?$

2) $I_{O \text{ ideal}}$ y $I_{O \text{ real}}$ si: $R_D = 10 k$

$$\rightarrow V_{DS} = V_{DD} - I R$$

$$\bullet V_{DS1} = V_{DD} - I_{ref} R$$

$$\bullet V_{DS2} = V_{DD} - I_O \cdot R_D$$